Cognition in the Age of AI (85-414/814)
Optional readings and other resources:

These materials are not required, but may be useful in gaining some background knowledge for this course, and a deeper dive into some advanced topics.

Getting started with Colab and PyTorch:

- More about Google Colab:
    - https://colab.research.google.com/notebooks/basic_features_overview.ipynb
- More tutorials in PyTorch:
    - https://docs.pytorch.org/tutorials/beginner/basics/intro.html
    - https://pytorch.org/tutorials/beginner/nn_tutorial.html
    - https://pytorch.org/tutorials/beginner/pytorch_with_examples.html

Other neural network coding tutorials:

- Neuromatch academy offers online courses in topics related to this class:
    - Computational Neuroscience
    - Deep Learning
    - NeuroAI
- The materials from these courses are all openly available (as Colab notebooks). You can check these out for further advanced exploration of course topics.

Introduction to neural network concepts for beginners:

- Intro to neural networks:
    - https://victorzhou.com/blog/intro-to-neural-networks/

- Intro to convolutional neural networks:
    - https://www.pinecone.io/learn/series/image-search/cnn/
    - https://towardsdatascience.com/convolutional-neural-networks-for-beginners-c1de55eee2b2/
    - https://colah.github.io/posts/2014-07-Understanding-Convolutions/

- Intro to language models:
    - https://mark-riedl.medium.com/a-very-gentle-introduction-to-large-language-models-without-the-hype-5f67941fa59e
    - https://developers.google.com/machine-learning/crash-course/llm
    - https://huggingface.co/learn/llm-course/chapter1/1

- Intro to self-attention and transformers
  - Illustrated Transformer: https://jalammar.github.io/illustrated-transformer/
  - https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/

- Linear algebra:
  - A nice video series reviewing basic concepts in linear algebra (3Blue1Brown): https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

Blogposts exploring the inner workings of neural networks:

- Curve detectors in CNNs: https://distill.pub/2020/circuits/curve-detectors/
- CNN feature visualization: https://distill.pub/2017/feature-visualization/
- Interpreting LLM features (Claude): https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html
- Multimodal neurons in LLMs: https://distill.pub/2021/multimodal-neurons/
- Superposition and interpretability: https://transformer-circuits.pub/2022/toy_model/index.html

Longer review articles that overview neural networks and deep learning:

This one is especially good for neuroscience/psychology background:
Kriegeskorte (2015), Deep Neural Networks: A New Framework for Modeling Biological Vision and Brain Information Processing.
https://doi.org/10.1146/annurev-vision-082114-035447

LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. Nature 521, 436–444 (2015).
https://doi.org/10.1038/nature14539

More theoretical discussion & critiques of deep networks for neuroscience:

Kanwisher, Khosla, & Dobs (2023). Using artificial neural networks to ask 'why' questions of minds and brains. https://doi.org/10.1016/j.tins.2022.12.008

Saxe, Nelli, & Summerfield (2021). If deep learning is the answer, what is the question?
https://doi.org/10.1038/s41583-020-00395-8

Richards et al (2019) A deep learning framework for neuroscience
https://doi.org/10.1038/s41593-019-0520-2

Guest & Martin (2023). On Logical Inference over Brains, Behaviour, and Artificial Neural Networks. https://doi.org/10.1007/s42113-022-00166-x

Bowers et al. (2022). Deep problems with neural network models of human vision. https://doi.org/10.1017/s0140525x22002813

Learn more about functional magnetic resonance imaging (fMRI):

- fMRI 4 Newbies – by Jody Culham (Western University)
    - https://www.fmri4newbies.com/lectures
- UCSD CFMRI:
    - https://cfmriweb.ucsd.edu/Research/whatisfmri.html
- Details of MR physics etc:
    - http://mriquestions.com/index.html

Learn more about the human visual system:

- DiCarlo, J. J., & Cox, D. D. (2007). Untangling invariant object recognition. Trends in cognitive sciences, 11(8), 333–341. https://doi.org/10.1016/j.tics.2007.06.010

- Grill-Spector, K., Weiner, K (2014). The functional architecture of the ventral temporal cortex and its role in categorization. Nat Rev Neurosci 15, 536–548. https://doi.org/10.1038/nrn3747

- Entry by Matteo Carandini about early visual processing:
    - http://www.scholarpedia.org/article/Area_V1

- Lecture notes from David Heeger at NYU (these are shared for several topics):
    - https://www.cns.nyu.edu/~david/courses/perception/lecturenotes/V1/lgn-V1.html